

PATENT A

Self-Sustaining Neural-Motor Energy Harvesting Loop for Autonomous Cognitive Systems

Drawing Sheets: 8 figures

FIG. 1 — System Architecture Block Diagram (8-Layer Loop)

FIG. 2 — Energy Balance Comparison

FIG. 3 — SNN Architecture (LIF Neuron Model)

FIG. 4 — Energy Harvesting Circuit

FIG. 5 — Activity-Dependent Energy Dynamics

FIG. 6 — Hardware Reference Design

FIG. 7 — Validation Results Summary

FIG. 8 — Energy-Bounded Recursive Control Architecture

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed February 2, 2026

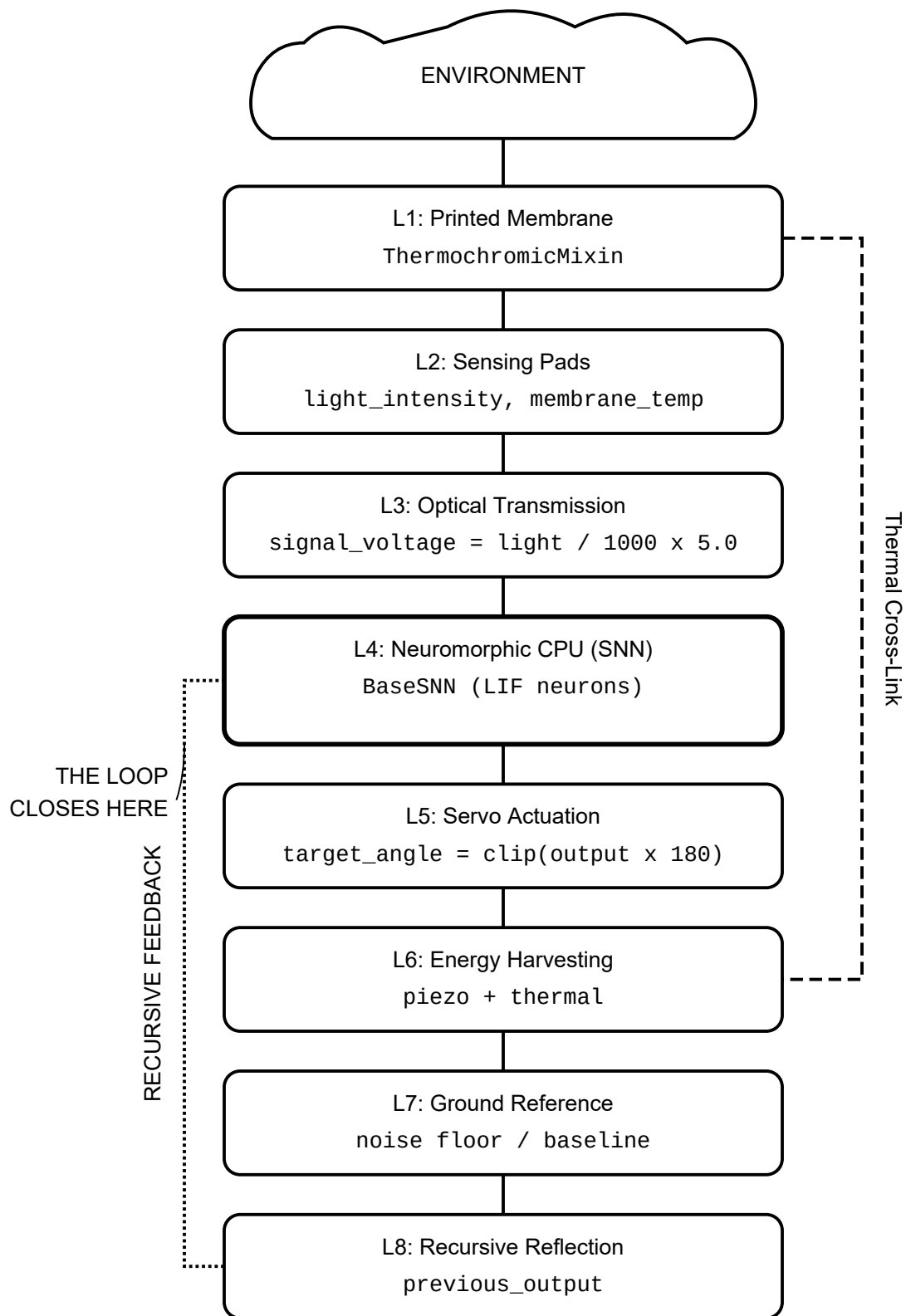


FIG. 1

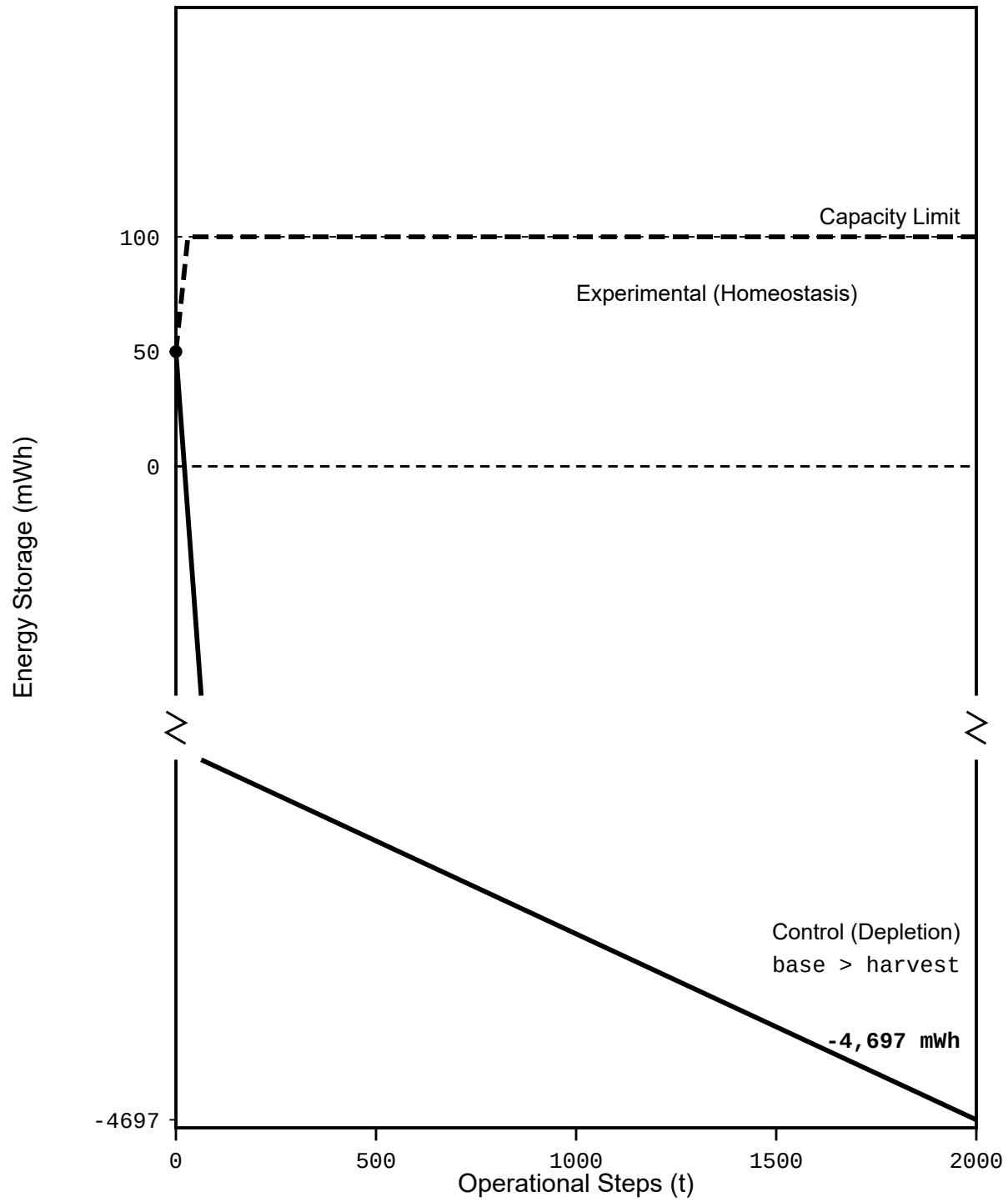
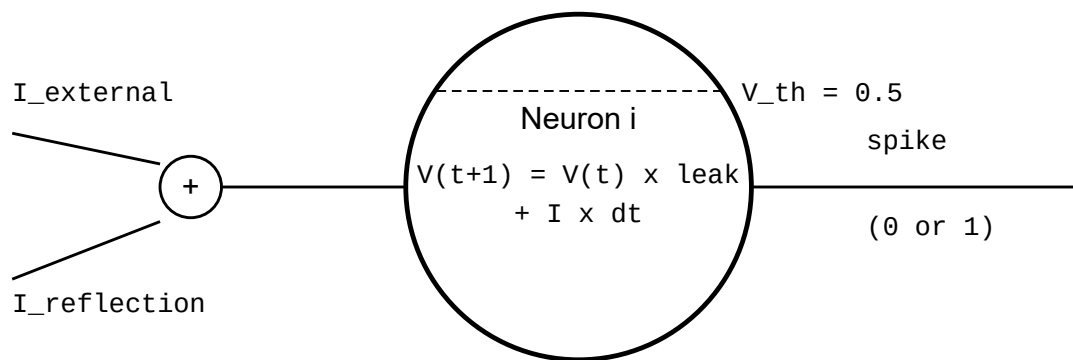
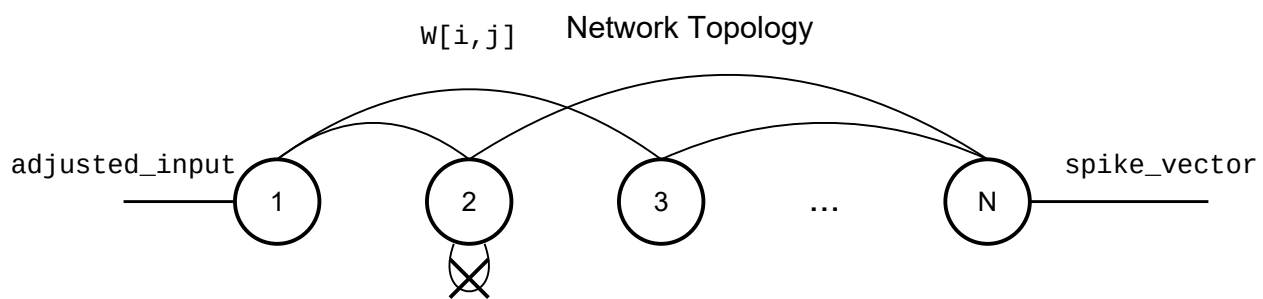


FIG. 2

Single LIF Neuron Detail



If $V \geq \text{threshold}$: spike, $V = 0$
 Refractory period: 2 steps



$W[i,i] = 0$ (no self-connections)
 $N = 50$ (operational), $N = 20$ (control)

STDP Learning Rule

Post fires: $W += A+ \times \text{pre_trace}$ (LTP)

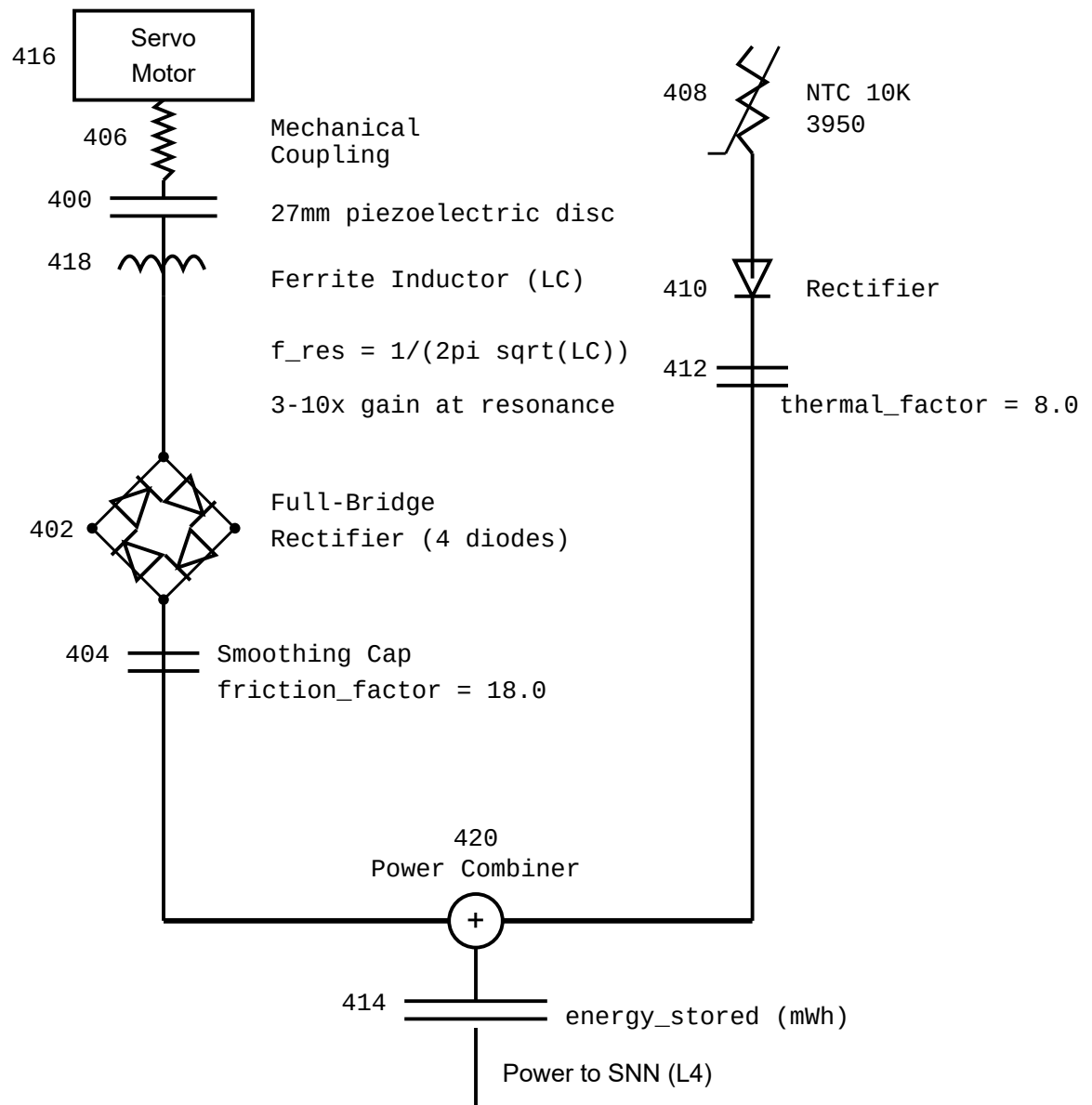
Pre fires: $W -= A- \times \text{post_trace}$ (LTD)

$A- > A+$ (stability bias)

FIG. 3

Piezoelectric Harvester

Thermoelectric Harvester



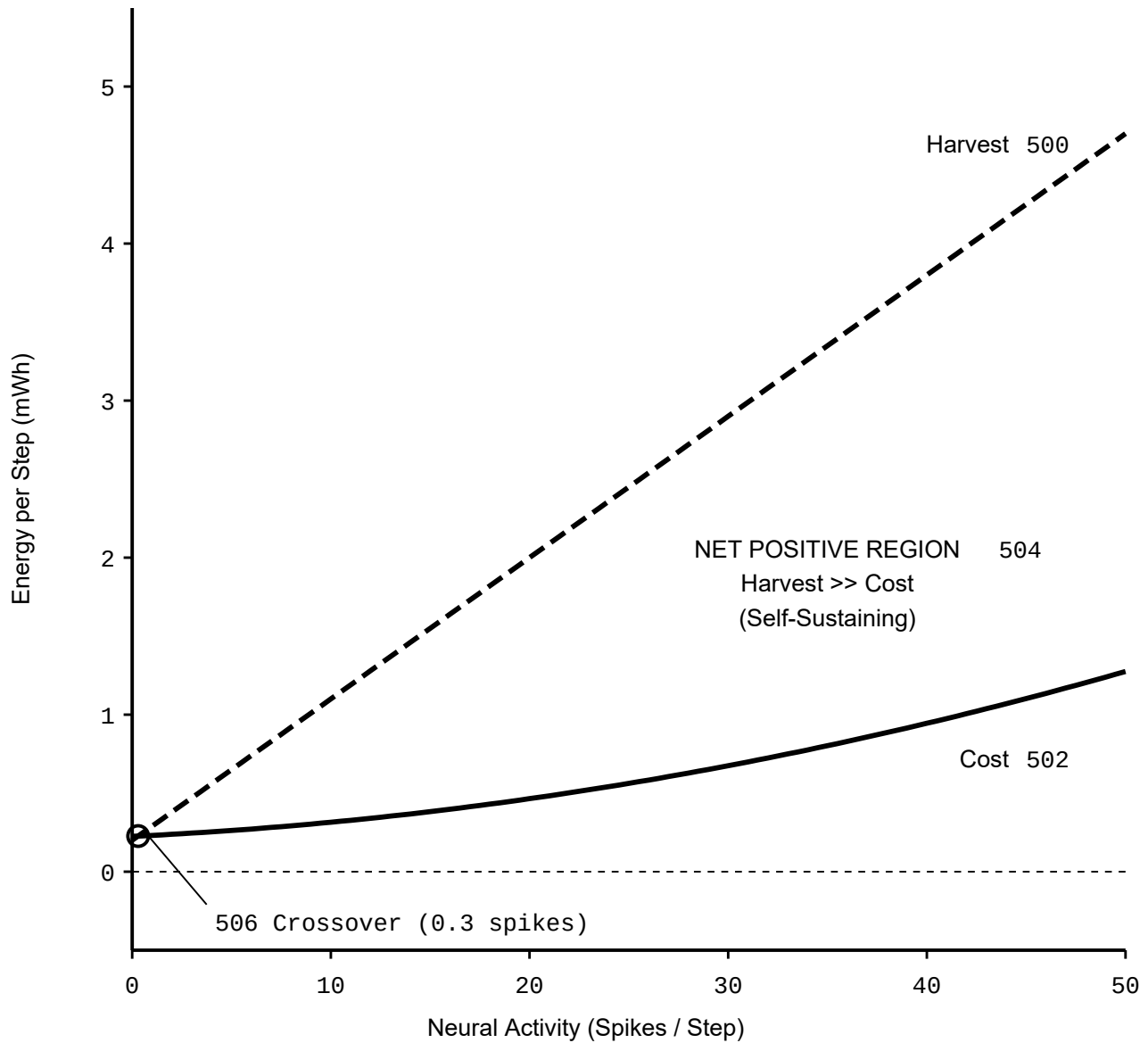
Power Budget

$$\text{Harvest} = \text{friction_harvest} + \text{thermal_harvest}$$

$$\text{Cost} = \text{base_consumption} + \text{activity_cost} \times \text{spike_count}^2$$

$$\text{Net} = \text{Harvest} - \text{Cost}$$

FIG. 4



$$\begin{aligned} \text{Harvest} &= 0.20 + (0.09 \times \text{spikes}) \text{ mWh/step} & 508 \\ \text{Cost} &= 0.225 + (0.006 \times s) + (0.0003 \times s^2) \text{ mWh/step} \\ \text{Net} &= \text{Harvest} - \text{Cost} \end{aligned}$$

Zone 1: Quiescent Drain
0 - 1 Spikes: Cost > Harvest

FIG. 5

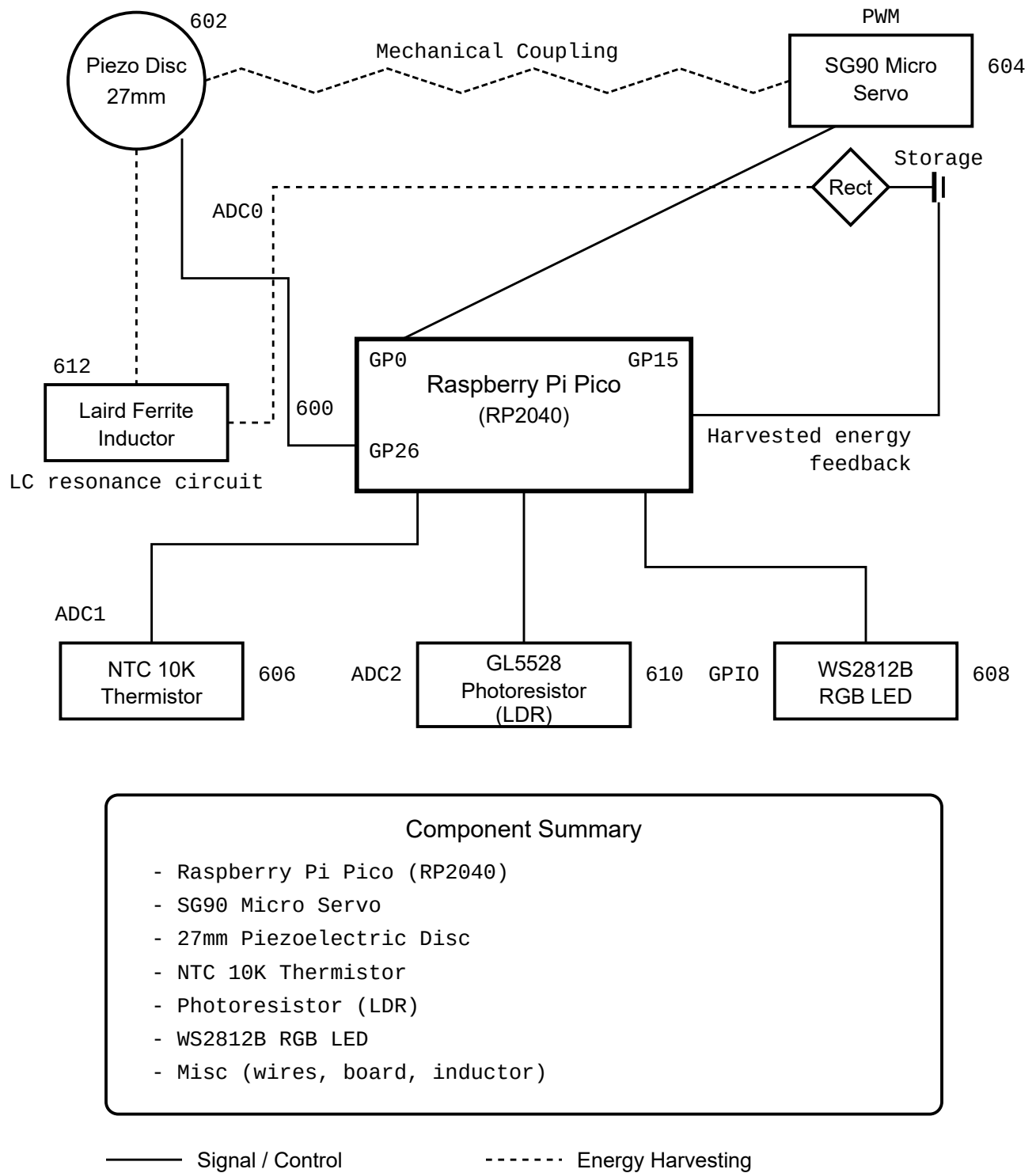
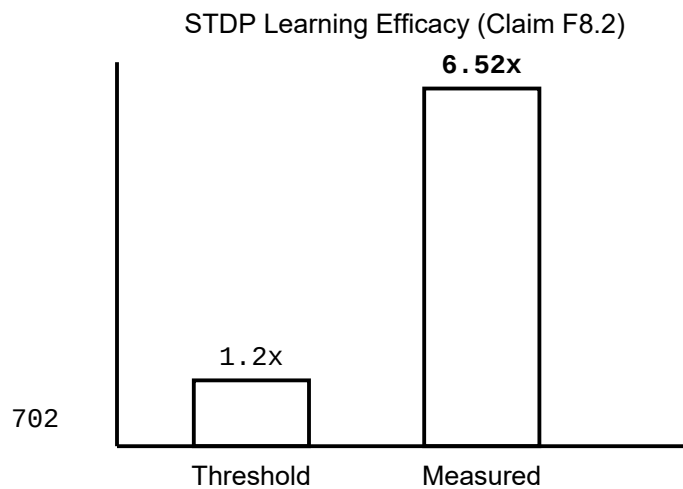


FIG. 6

700

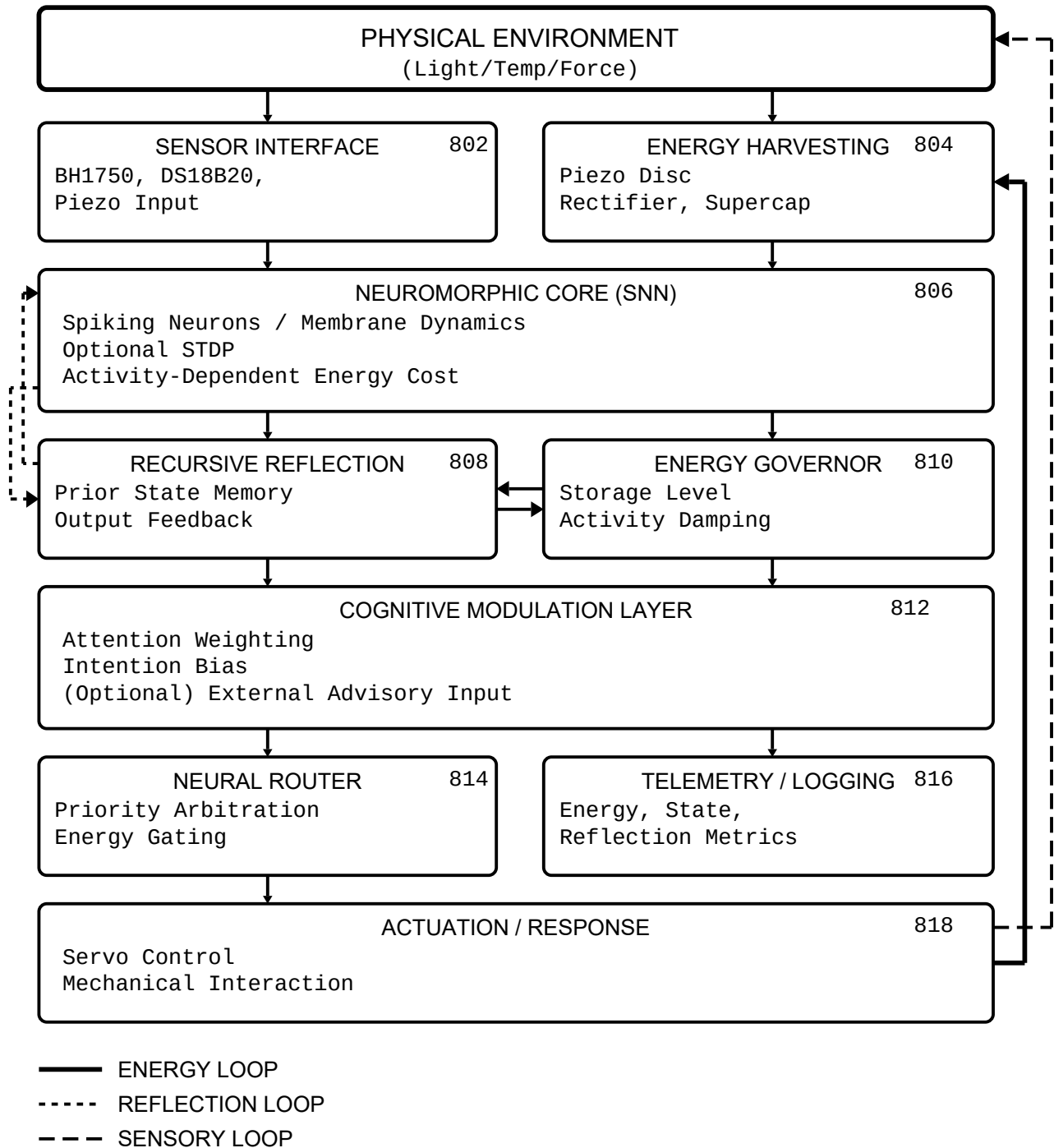
VALIDATION RESULTS SUMMARY

Claim	Criterion	Measured	Threshold	Result
Phase 7 Control Baseline (EnergyConfig)				
A	Closed-loop continuity	2000 pts	2000	PASS
B	Boundedness	bounded	bounded	PASS
C	Robustness (noise)	Std(theta) ≤ 45	≤ 45	PASS
D	Input-output gain	corr ≥ 0.2	≥ 0.2	PASS
E	Saturation ratio	sat ≤ 0.20	≤ 0.20	PASS
Phase 7 Full Integration (Control confirms energy depletion)				
--	Energy at 2000 steps	-4,697 mWh	< 0	CONFIRMED
MCP Integration (BalancedEnergyConfig)				
--	Energy at 2000 steps	100.0 mWh (capped)	> 0	CONFIRMED
Phase 8 STDP				
F8.1	Weight entropy decrease	2.95 to 2.02 bits	decrease	PASS
F8.2	STDP MI $> 1.2x$ frozen	6.52x	> 1.20	PASS
F8.3	Weight convergence	0.00%	$< 10\%$	PASS



All claims validated. System is reproducible via git tags.

FIG. 7



All computation is constrained by real-time energy availability derived from mechanical interaction.

FIG. 8

PATENT B

Configurable Recursive Self-Observation Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient Modulation

Drawing Sheets: 6 figures

FIG. 1 — Self-Observation Feedback Loop

FIG. 2 — Reflection Coefficient Spectrum

FIG. 3 — Dynamic Modulation Sources

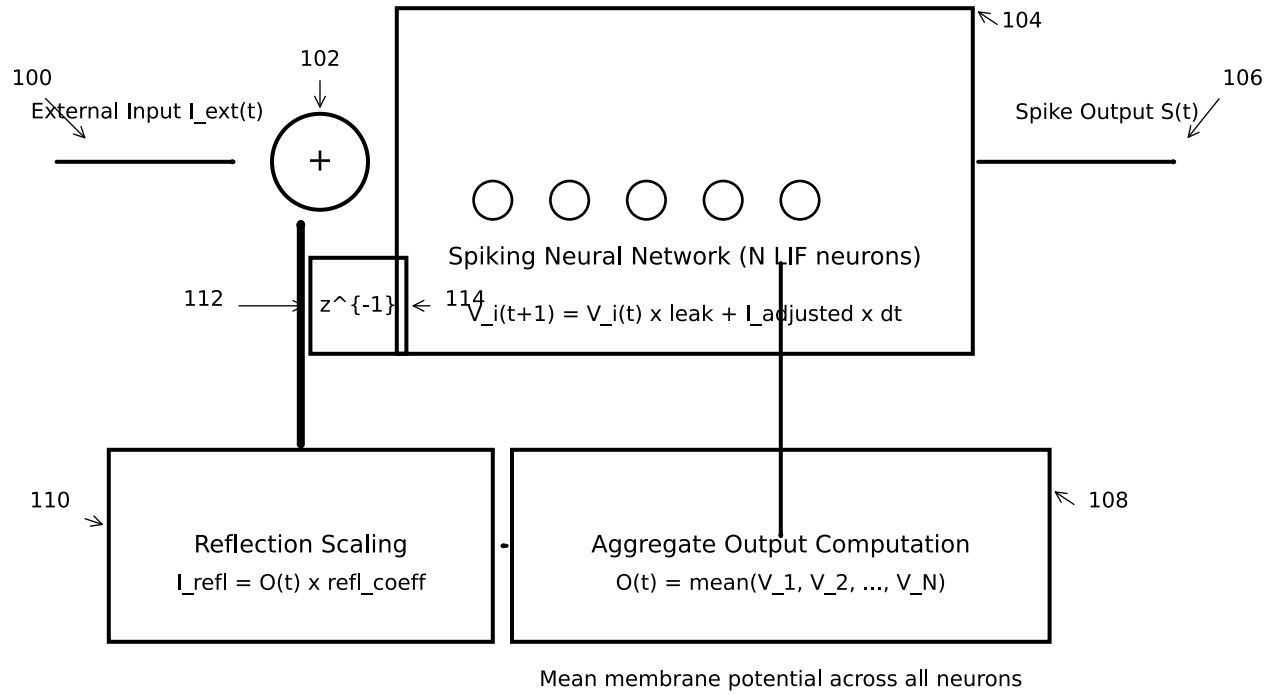
FIG. 4 — Self-Referential Learning Loop (STDP + Self-Observation)

FIG. 5 — Energy-Aware Self-Observation Regulation

FIG. 6 — End-to-End Signal Flow with Self-Observation Integration

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed February 2, 2026



Self-observation: the network observes its own prior aggregate state

$$I_{adjusted}(t+1) = I_{ext}(t+1) + I_{reflection}(t)$$

FIG. 1

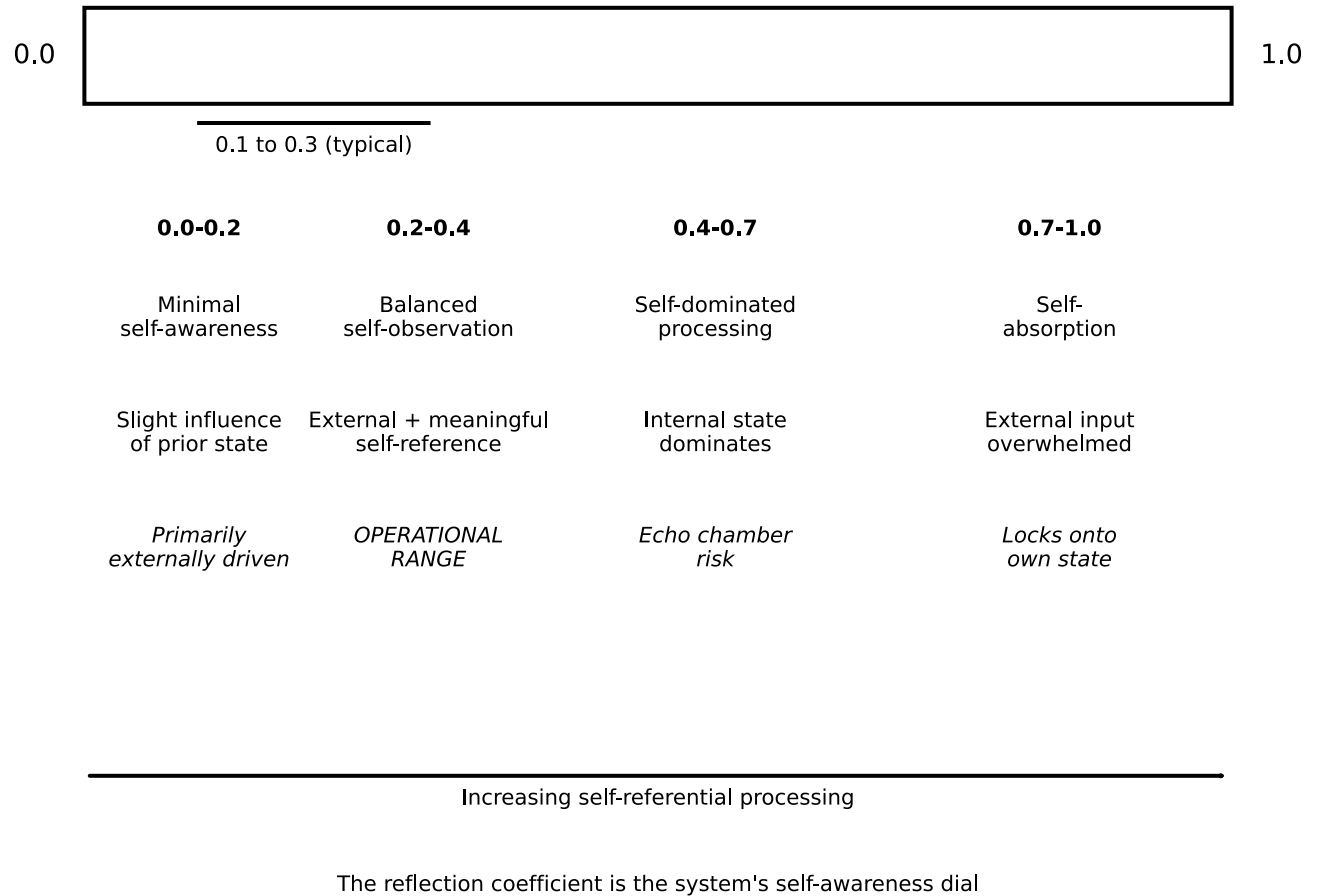


FIG. 2

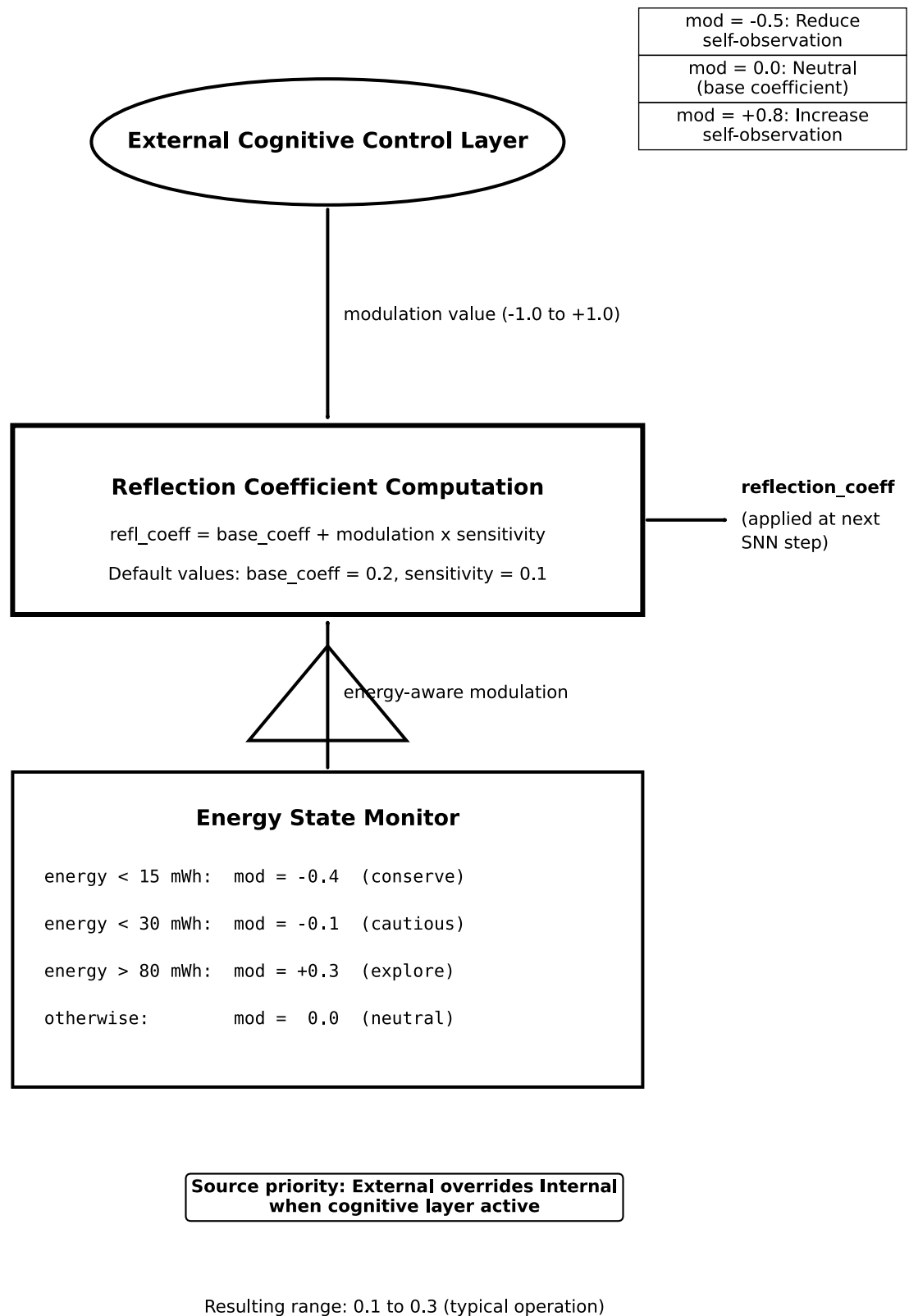


FIG. 3

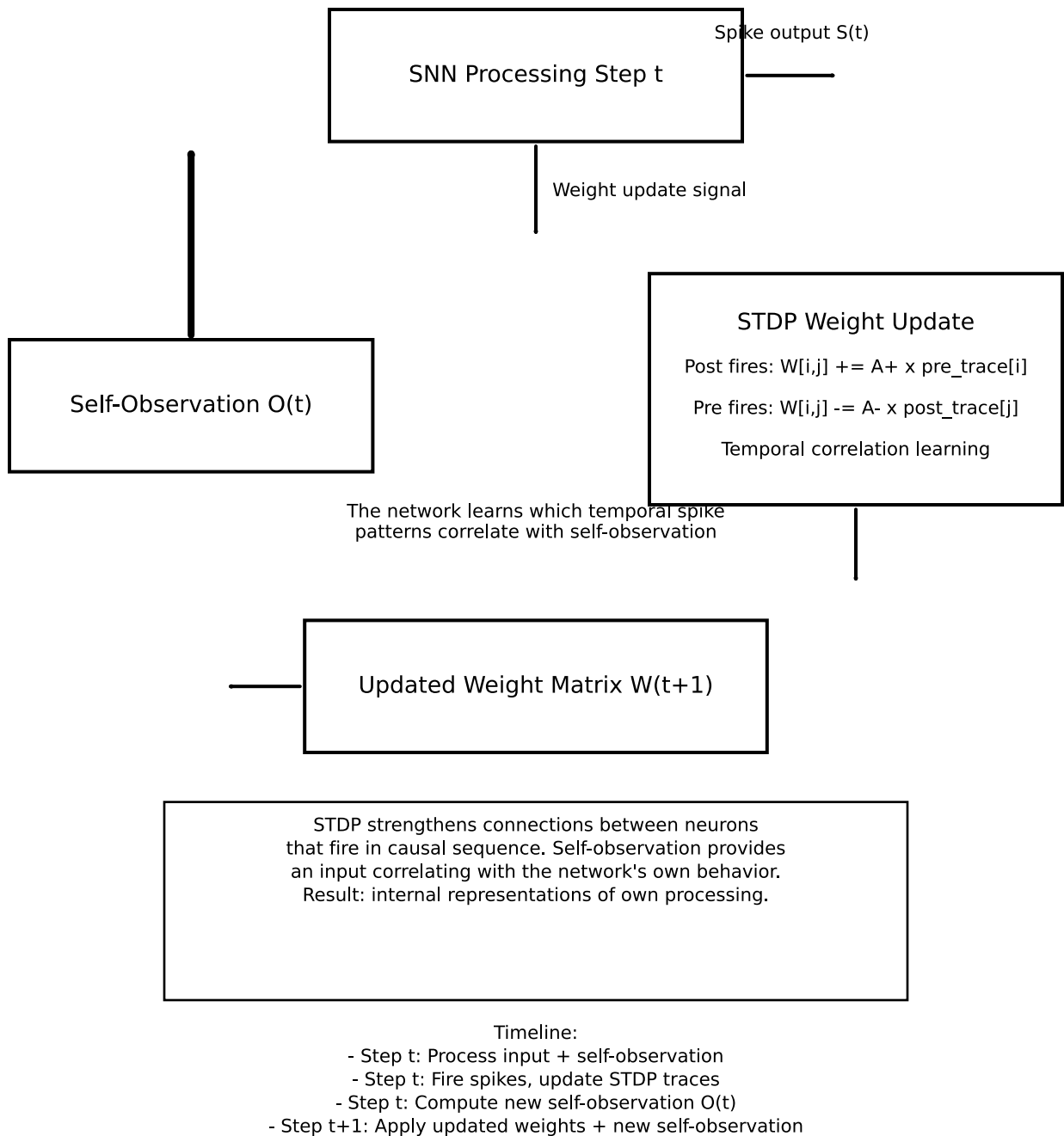
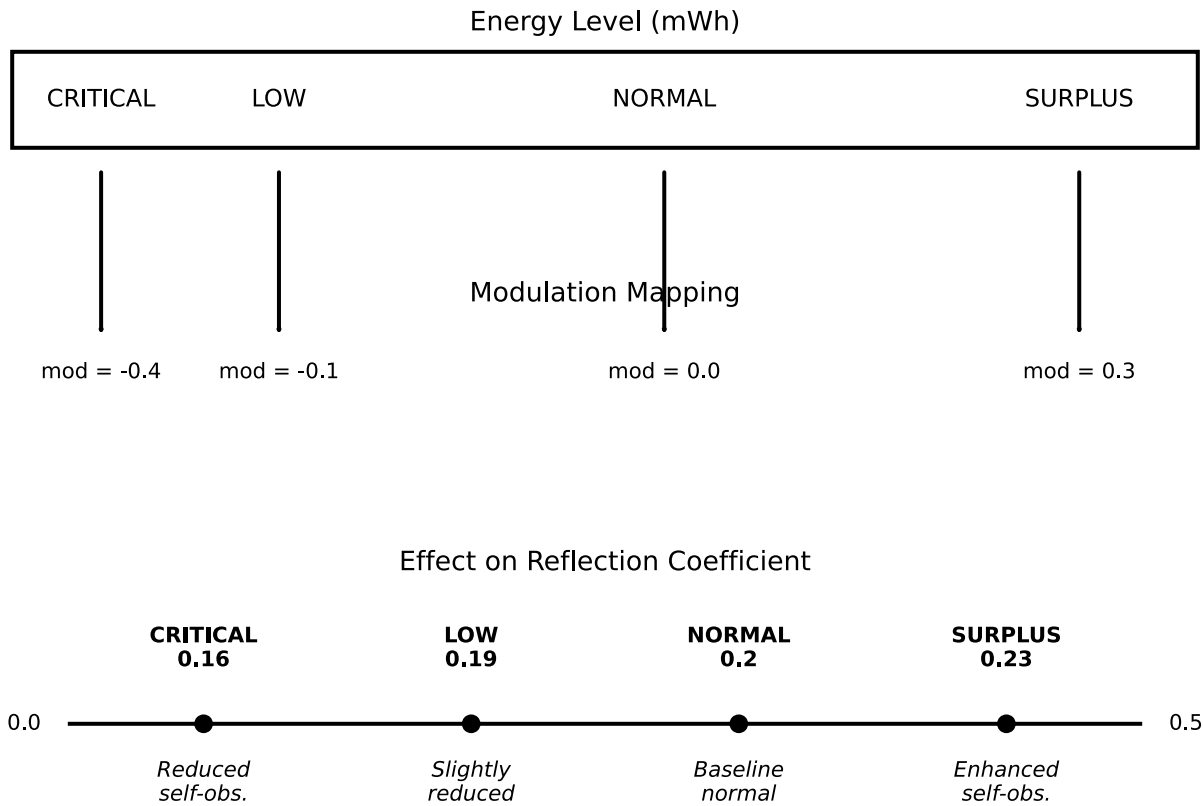


FIG. 4



Energy-aware regulation ensures the system reduces
cognitive overhead when energy is scarce and explores
deeper self-reference when energy is abundant

FIG. 5

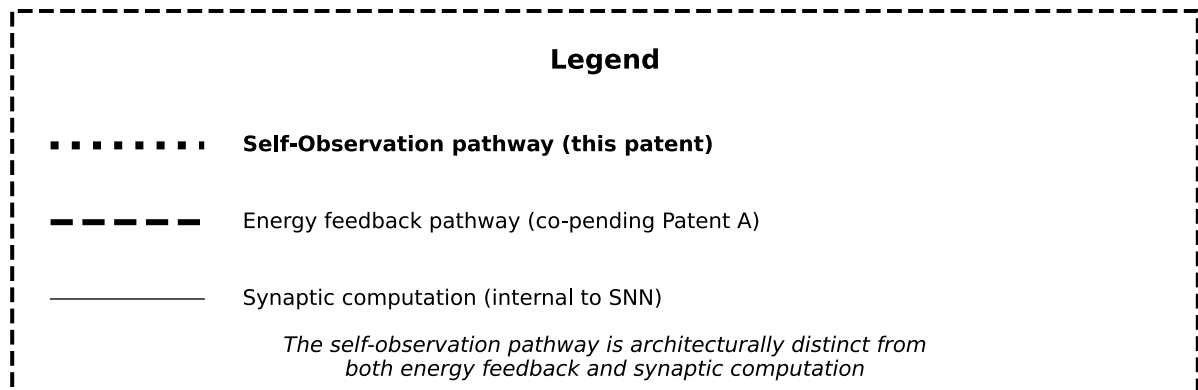
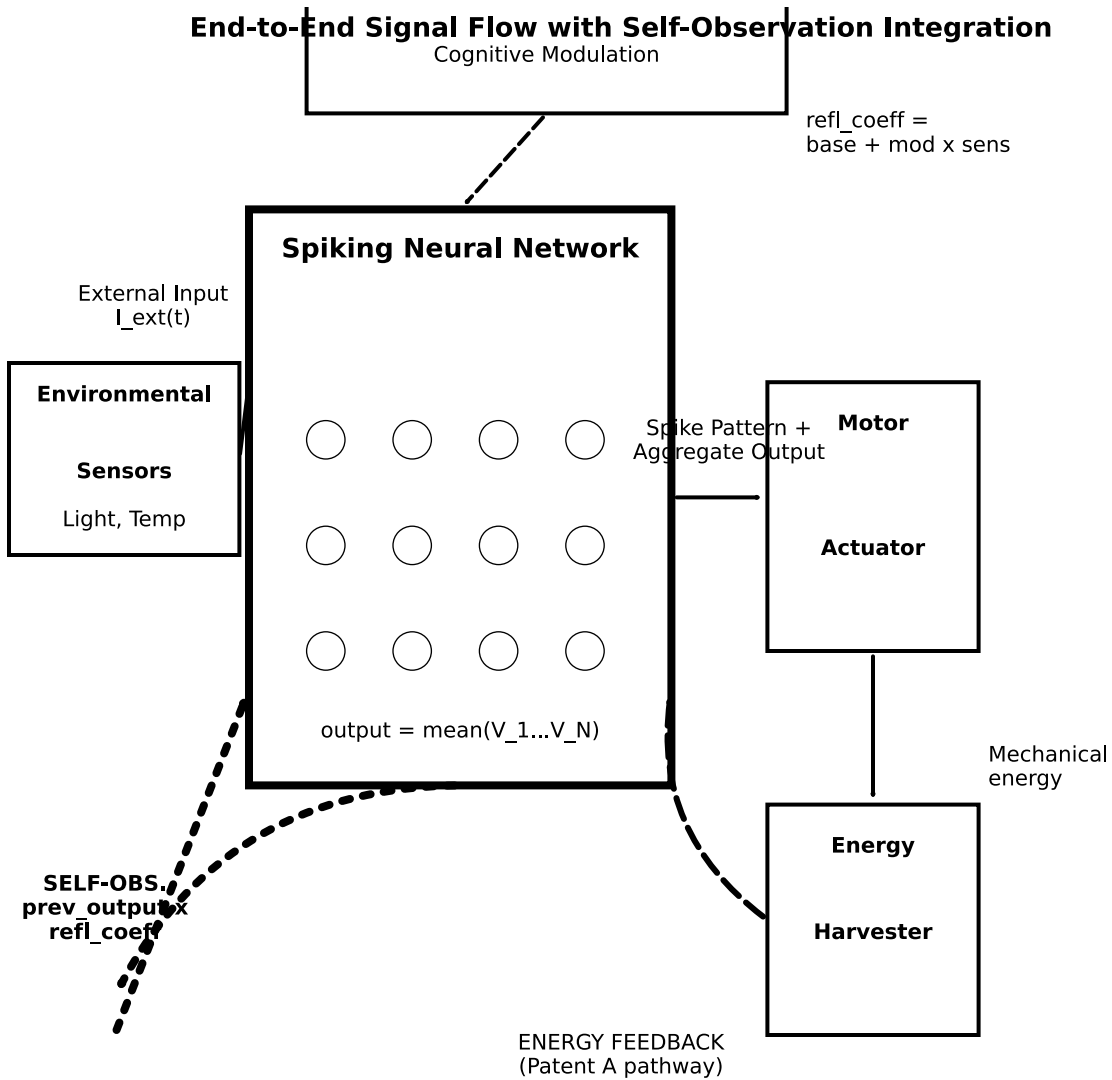


FIG. 6

PATENT C

Method and System for Autonomous Self-Regulation and Cognitive Resynchronization in Neural Processing Systems During Disconnection from External Control Layers

Drawing Sheets: 7 figures

FIG. 1 — System Architecture with Fallback

FIG. 2 — State Transition Diagram

FIG. 3 — Energy-Aware Modulation Curve

FIG. 4 — Autonomous Input Generator Output

FIG. 5 — Resynchronization Payload Structure

FIG. 6 — Recovery Timeline Diagrams

FIG. 7 — End-to-End Signal Flow: Connected vs. Autonomous

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed February 2, 2026

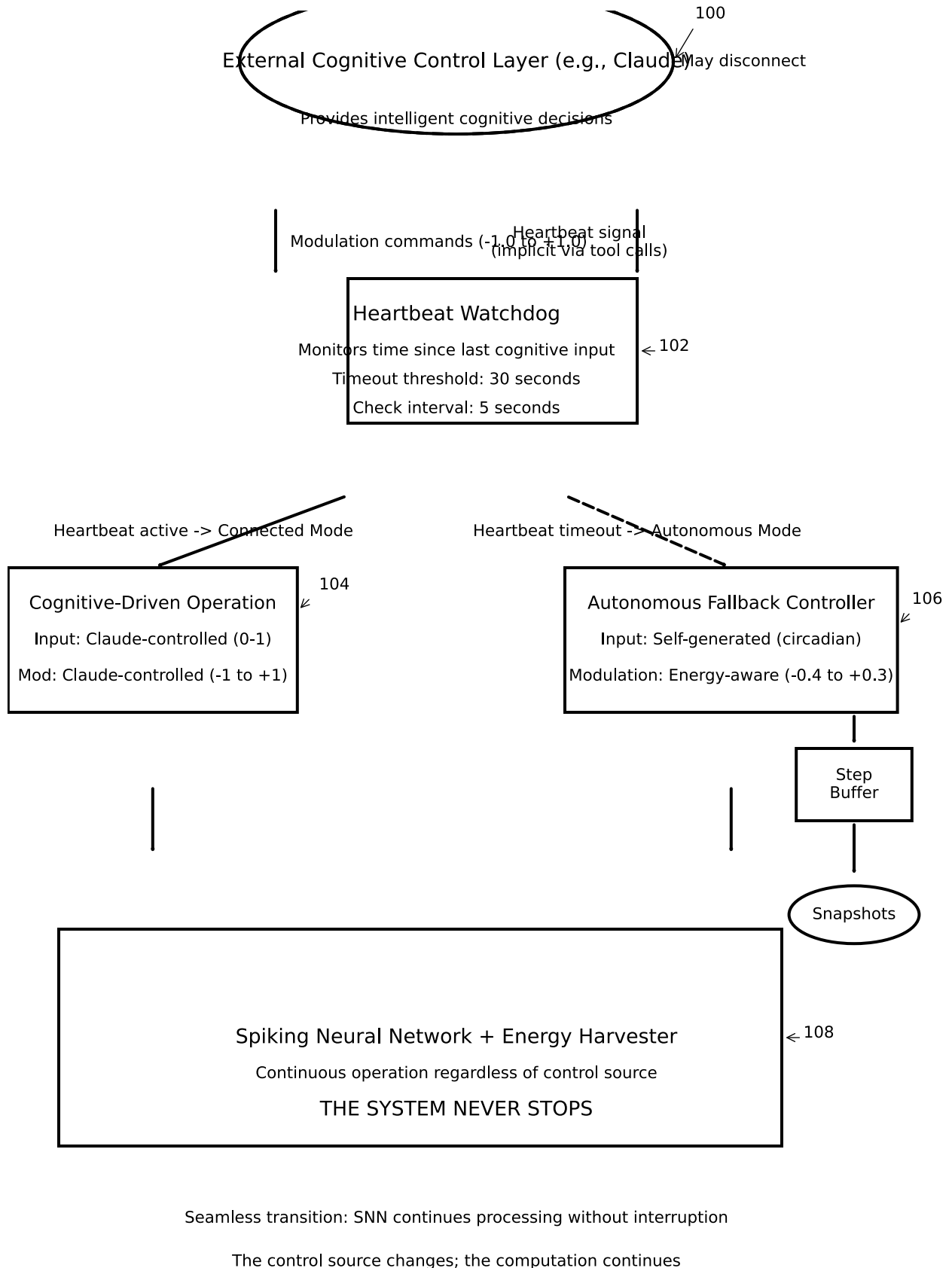


FIG. 1

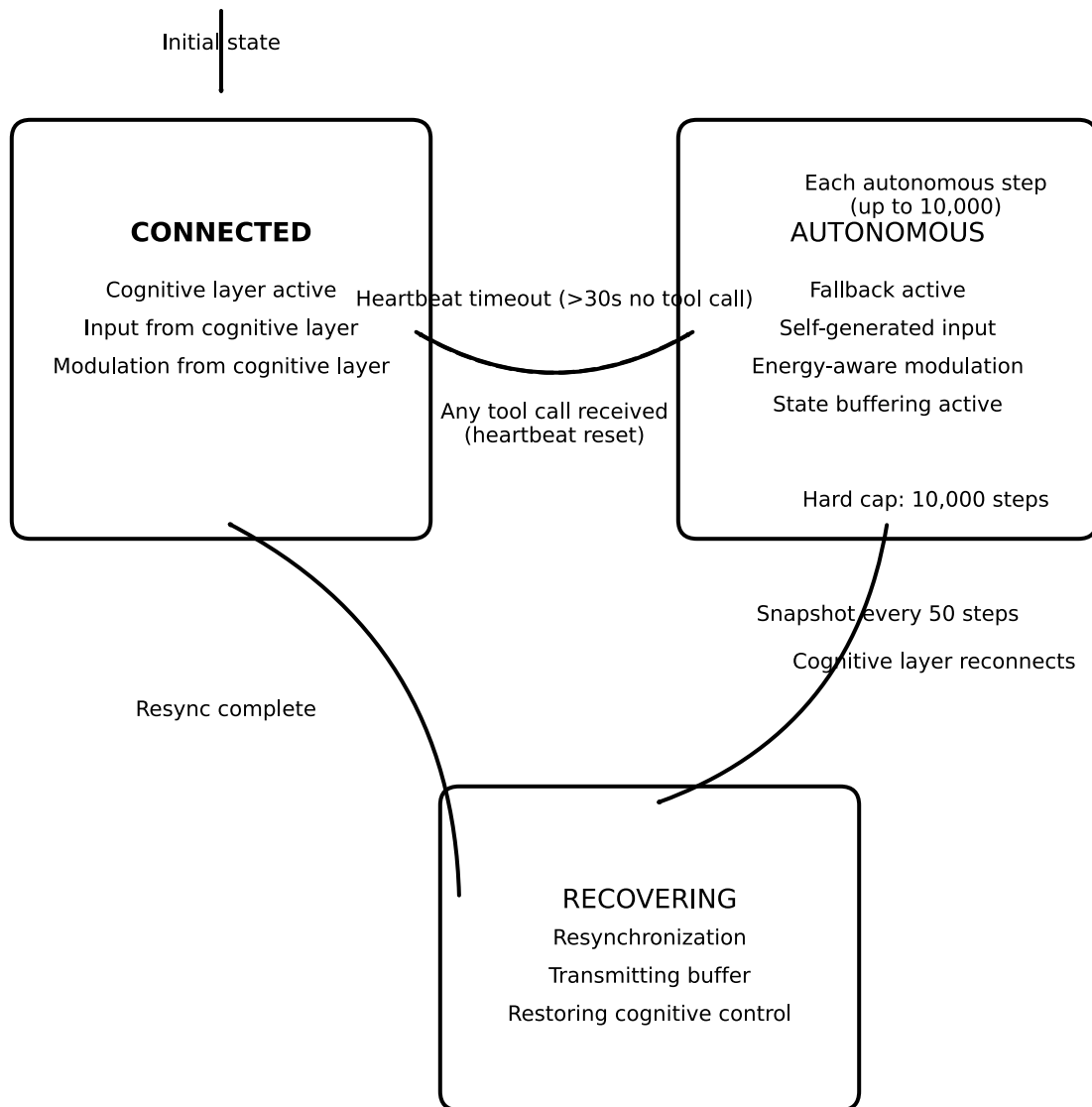
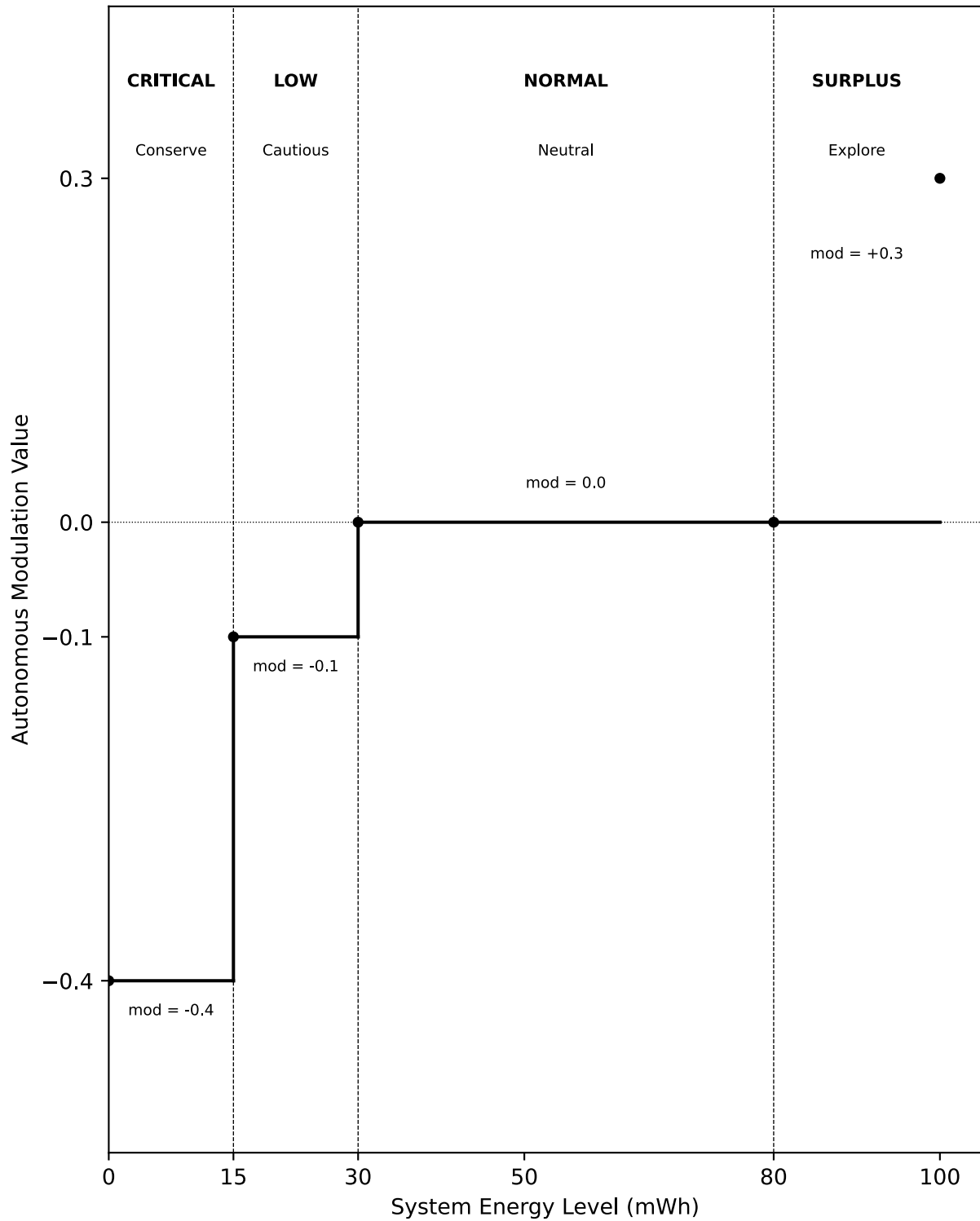


FIG. 2



System autonomously adjusts processing intensity based on energy

FIG. 3

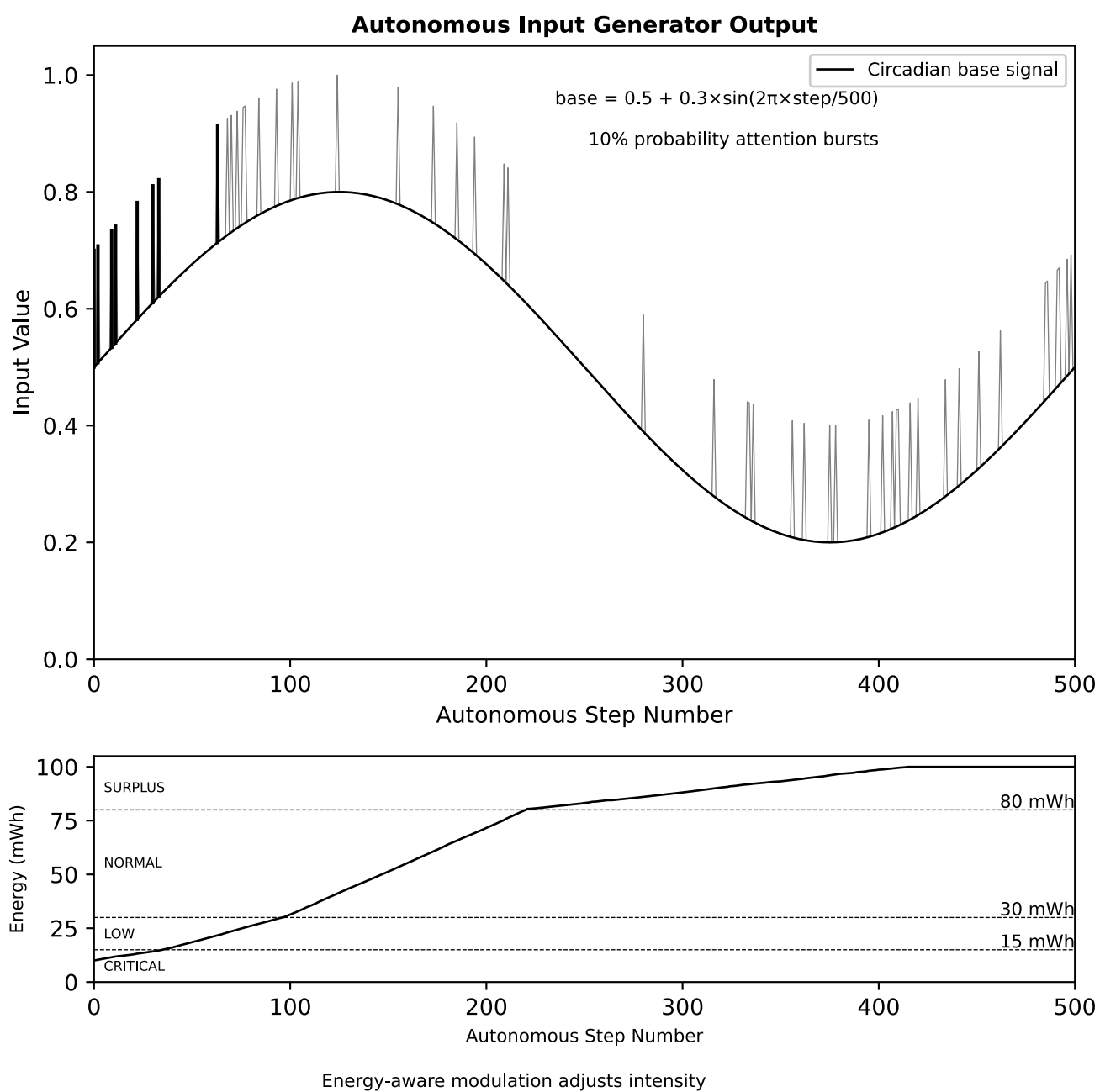


FIG. 4

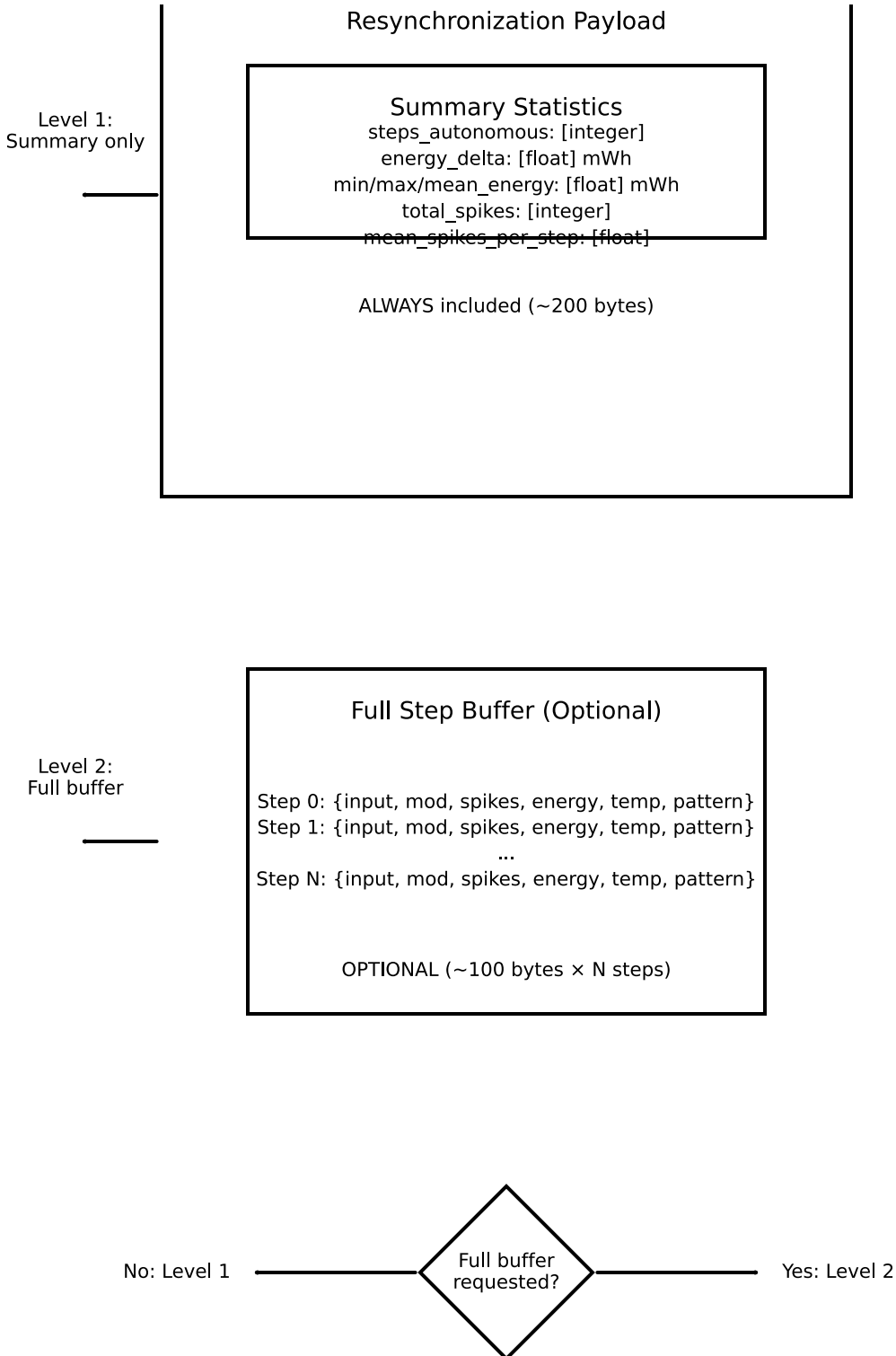


FIG. 5

Timeline A — Normal Reconnection:

Connected	Timeout	Autonomous	Resync	Connected
[Claude active]	[Watchdog]	[Self-regulated]	[Buffer sent]	[Restored]

Timeline B — Crash Recovery:

Connected	CRASH	Restart	Snapshot	Connected
[Normal op]	[Server dies]	[Relaunched]	[latest.json]	[Restored]

Max state loss = 50 steps (snapshot interval)

Timeline C — Clean Shutdown:

Connected	Autonomous	EOF Signal	Shutdown
[Normal op]	[Fallback]	[stdin closed]	[Snapshot written]

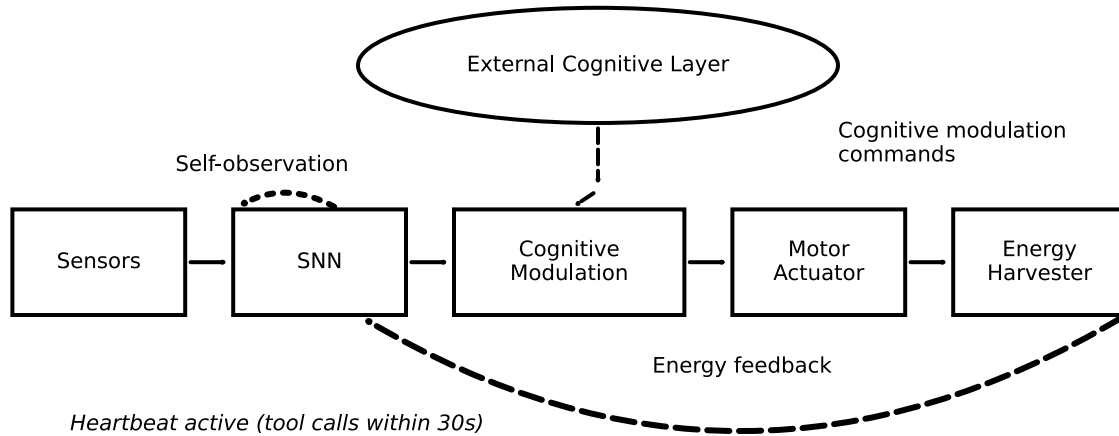
State preserved for next session — zero data loss

Solid segments: System actively processing

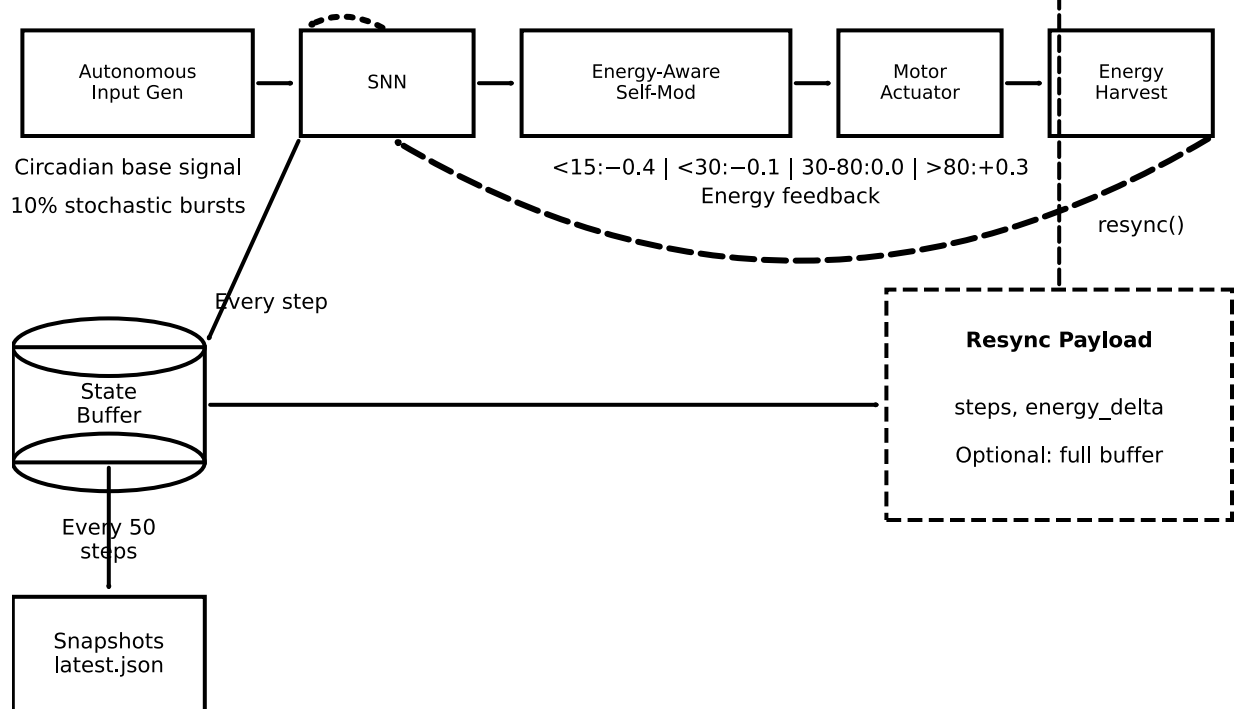
Dashed segments: System in transition

Bold marker at each state change: No data loss at any transition

FIG. 6

CONNECTED MODE**End-to-End Signal Flow: Connected vs. Autonomous Operation****DISCONNECTION EVENT (heartbeat timeout > 30s)**

Seamless transition — no interruption to SNN or Harvester

AUTONOMOUS MODE**Fallback changes INPUT SOURCE and MODULATION SOURCE**

Core neural processing, motor actuation, energy harvesting

continue WITHOUT INTERRUPTION through the transition

FIG. 7 Forward signal — — — — — Energy feedback Self-observation